

HOGFJALLSHOTELLET, SwedenWMO: **023110**Lat: **61.1583N**Lon: **13.1138E**Elev: **2464**StdP: **13.43**Time Zone: **1.00 (EUC)**Period: **01-16**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.4	(c) 3.1	(d) -5.8	(e) 4.4	(f) 4.1	(g) -2.2	(h) 5.3	(i) 5.4	(j) 21.8	(k) 30.7	(l) 20.0	(m) 28.8	(n) 2.7	(o) 50	(p) 0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.8	(c) 72.0	(d) 57.4	(e) 68.8	(f) 55.8	(g) 65.6	(h) 54.4	(i) 60.0	(j) 67.8	(k) 58.1	(l) 64.9	(m) 56.5	(n) 62.4	(o) 7.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.1	76.1	62.1	55.4	71.4	60.2	53.8	67.3	59.0	27.5	67.7	26.2	65.1	25.1	62.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.7	15.3	-4.6	76.4	4.8	3.0	-8.0	78.6	-10.9	80.3	-13.6	82.0	-17.1	84.2
			-4.8	61.5	4.2	3.1	-7.9	63.7	-10.3	65.5	-12.7	67.3	-15.7	69.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	20.0	20.5	26.0	34.7	43.3	51.5	55.9	53.4	45.3	35.7	28.0	22.3
	DBStd	14.46	7.85	7.94	7.59	6.42	6.95	6.74	5.45	5.21	5.57	6.85	7.50	8.67
	HDD50	5398	929	826	743	458	232	56	8	24	160	443	660	859
	HDD65	10416	1394	1246	1208	908	673	408	288	359	591	907	1110	1324
	CDD50	466	0	0	0	0	25	102	190	130	18	1	0	0
	CDD65	8	0	0	0	0	0	3	5	0	0	0	0	0
	CDH74	30	0	0	0	0	0	12	18	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.4	6.3	6.5	8.0	7.8	7.7	7.4	7.2	7.0	8.2	8.2	7.7	6.4
Precipitation	PrecAvg	29.3	1.9	1.3	1.2	1.7	2.4	3.3	3.5	4.0	2.8	2.8	2.4	1.9
	PrecMax	40.5	4.4	2.2	2.7	4.4	6.3	6.5	6.3	6.3	5.0	6.5	6.9	3.5
	PrecMin	24.3	0.4	0.2	0.2	0.2	0.4	0.2	0.8	1.5	0.8	0.5	0.7	0.2
	PrecStd	3.6	1.0	0.6	0.6	1.2	1.3	1.4	1.2	1.4	1.2	1.5	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	38.4	47.5	56.2	69.6	75.5	76.3	71.5	63.0	51.5	46.5	40.6
		MCWB	33.6	32.9	39.4	46.8	53.7	59.0	61.0	56.4	52.4	45.4	43.6	36.7
	2%	DB	34.0	35.3	42.8	52.0	63.5	71.9	72.6	68.4	59.0	48.9	42.6	37.5
		MCWB	32.2	32.7	35.4	43.1	50.3	56.6	58.9	55.4	50.9	45.8	41.0	35.0
	5%	DB	32.2	33.0	39.4	48.1	59.3	67.9	69.3	65.1	55.9	46.7	40.1	35.6
		MCWB	30.9	30.9	34.1	40.9	47.8	53.9	57.2	54.7	49.7	44.0	38.7	33.5
	10%	DB	30.8	31.1	36.3	44.9	54.5	63.4	65.9	61.8	53.2	45.0	37.7	33.4
		MCWB	29.4	29.4	32.3	38.5	45.2	52.3	55.7	53.9	48.4	42.9	36.6	31.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	35.5	39.6	47.8	55.7	60.9	63.8	61.0	55.2	49.4	44.3	38.7
		MCDB	35.0	36.6	46.0	54.0	64.6	71.7	72.9	65.8	60.7	49.9	45.3	39.3
	2%	WB	32.8	33.2	37.1	44.5	52.3	58.4	60.8	58.7	52.6	46.9	41.3	35.8
		MCDB	33.3	34.3	41.5	50.3	60.2	68.0	69.3	63.5	56.4	47.9	42.5	36.7
	5%	WB	31.4	31.3	34.7	41.7	49.5	56.0	58.7	57.0	50.9	44.8	39.0	33.9
		MCDB	32.2	32.7	38.0	46.5	56.7	64.8	66.3	60.8	54.2	45.8	40.1	35.0
	10%	WB	29.5	29.4	32.7	39.0	47.0	53.3	57.2	55.2	49.4	43.0	36.7	31.8
		MCDB	30.6	30.9	35.6	43.6	52.8	61.6	63.8	59.9	52.3	44.6	37.7	32.9
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.7	10.0	11.2	13.2	14.0	12.8	11.2	10.5	8.2	7.0	7.7
		MCDBR	7.3	9.4	12.9	15.6	18.6	18.4	17.5	15.3	13.3	10.2	9.2	9.1
		MCWBR	7.3	8.3	8.5	9.4	9.9	9.2	8.0	7.0	7.9	7.7	8.4	8.3
	5% WB	MCDBR	7.2	8.6	11.2	15.5	16.7	16.7	15.0	12.0	11.3	8.4	8.5	8.6
		MCWBR	7.6	8.3	8.4	10.0	9.9	9.6	8.1	7.1	8.1	7.7	8.3	8.6
Clear-Sky Solar Irradiance	taub	0.212	0.258	0.284	0.320	0.326	0.320	0.337	0.328	0.299	0.285	0.230	N/A	
	taud	2.167	2.297	2.317	2.388	2.480	2.519	2.476	2.493	2.564	2.511	2.251	N/A	
	Ebn at Noon	178	237	268	279	285	289	280	274	265	228	167	N/A	
	Edh at Noon	11	19	27	30	30	30	30	27	21	16	10	N/A	
All-Sky Solar Radiation	RadAvg	90	299	788	1273	1490	1667	1503	1172	768	344	107	50	
	RadStd	12	58	81	135	144	129	203	112	94	62	29	6	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page